Kron-Reduced vs. Full Graph Laplacians for Water-Quality Regression on the Colorado River Network

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Abstract

The graph-Laplacian water-quality models for the Colorado basin use a Kron (Schur-complement) reduction of the 67,401-node river network onto the 26 nodes with water-quality (WQ) stations. We test whether this reduction changes the model. For five MERIT-based edge weightings, on both the uncompressed and the compressed (67-node) graph, the Kron and full Laplacians agree to within 10^{-8} on every quantity the reduction should preserve. Re-fitting the nested leave-one-node-out (LOO) regression with the full Laplacian changes R^2 by at most 10^{-10} and selects identical hyperparameters in every fold. Kron reduction is therefore exact, and about $150\times$ faster. We also correct several problems in the original analysis. The most consequential is that feature sets chosen by looking at outer-LOO results overstated performance: by about 0.13 in R^2 for specific conductance and 0.02–0.08 for DO, pH and turbidity. With selection done inside the inner loop, the graph term clearly helps for pH and specific conductance, while seasonal and MERIT predictors matter much more for water temperature, DO and turbidity.

1 Setup

Data. Hourly USGS measurements at 28 stations, mapped to 26 river-network nodes. Depending on the target, 17–24 nodes have usable data (DO 17, water temperature 24, pH 19, specific conductance 24, turbidity 17). The network is the largest connected component of the MERIT-derived Colorado graph: 67,401 nodes and 67,400 edges, each edge about 70–120 m long.

Model. For node i and hour t, the prediction is $\hat{y}_{it} = \beta_0 + z_{it}^\top \beta + g_i$, where z_{it} are standardized predictors and g_i is a node offset. The fit minimizes

$$\sum_{i \in \text{train}} \frac{1}{n_i} \sum_t \big(y_{it} - \beta_0 - z_{it}^\top \beta - g_i\big)^2 \;+\; \lambda_\beta \sum_{k \geq 1} \beta_k^2 \;+\; \lambda\, g^\top \tilde{L}\, g, \qquad \text{s.t. } \textstyle\sum_{i \in \text{WQ}} g_i = 0, \tag{1}$$

with $g^\top L g = \sum_{(i,j)} w_{ij}(g_i - g_j)^2$ and $\tilde{L} = L/s$. The scale s is the mean positive diagonal of the Kron matrix. The weight $1/n_i$ makes each station count equally. A held-out node has no data term, so its offset is set by the smoothness penalty alone. The hyperparameters λ and λ_β (and, where stated, the edge weighting) are tuned by an inner LOO. Performance is the node-weighted R^2 of the outer-LOO predictions.

Edge weightings. $w_{ij} = \exp(\text{score}_{ij})/d_{ij}$, where d_{ij} is the edge length in km and the score mixes z-scored log upstream area, log width and log slope. There are five schemes: `distance_only`, `area`, `width`, `slope` and `area_width`.

Kron reduction. Partitioning nodes into WQ nodes B and the rest I,

$$L_{\text{kron}} = L_{BB} - L_{BI} L_{II}^{-1} L_{IB} \in \mathbb{R}^{26\times 26}. \tag{2}$$

The non-WQ nodes have no data term in (1), so minimizing over their g in closed form leaves exactly $g_B^\top L_{\text{kron}} g_B$. The two formulations should therefore be the same model. This report checks that numerically.

2 Are the Laplacians the same?

The Kron and full matrices have different sizes, so we compare what Kron reduction must preserve:

- **Effective resistance** between every pair of WQ nodes, computed on the full Laplacian by sparse solves and on L_{kron}.
- **Dirichlet energy:** for random WQ values x_B with harmonic extension x, $x^\top L x$ should equal $x_B^\top L_{\text{kron}} x_B$.
- **Compression:** Kron(uncompressed) vs Kron(compressed), compared directly.

As a reference for what a *wrong* reduction looks like, we also compare against the sub-matrix L_{BB}, i.e. simply deleting the non-WQ nodes.

Table 1: Largest relative errors, uncompressed graph. All values are at rounding-error level, except the deliberately wrong L_{BB} reference.

Weighting	Eff. resistance	Energy	Kron unc. vs comp.	Naive L_{BB} (reference)
`distance_only`	1.9×10^{-9}	2.4×10^{-12}	6.7×10^{-13}	205
`area`	4.7×10^{-9}	1.5×10^{-12}	5.1×10^{-13}	368
`width`	4.1×10^{-10}	3.1×10^{-12}	6.4×10^{-13}	253
`slope`	7.3×10^{-9}	8.8×10^{-12}	2.6×10^{-12}	257
`area_width`	6.0×10^{-9}	2.6×10^{-12}	8.1×10^{-13}	359

On the compressed graph, the resistance and energy errors are below 1.5×10^{-12} for every weighting. Compression is also exact. Each compressed edge replaces a chain of resistors in series, and series resistances add, so the total resistance between stations is unchanged.

3 Does Kron change the regression?

The nested LOO was re-run with four Laplacians, using exactly the same solver code:

Variant	Laplacian in (1)	Size
`kron_uncompressed`	Kron of the 67k graph (original method)	26×26
`kron_compressed`	Kron of the 67-node graph	26×26
`full_uncompressed`	full Laplacian, g on every node	$67{,}401 \times 67{,}401$
`full_compressed`	full Laplacian, g on every node	67×67

The full variants never form a Schur complement. They solve the complete (β, g) system by block elimination with a sparse factorization. Each full Laplacian is divided by the same s as its Kron reduction, so λ has the same meaning in all four.

Table 2: Outer-LOO R^2, chemistry predictors only. "Graph, $1/d$" is `distance_only`. The last column is the largest change in R^2 between `kron_uncompressed` and any other variant, over both graph modes.

Target	Linear	Graph, $1/d$	Graph, tuned weighting	Max $\lvert\Delta R^2\rvert$
DO concentration	0.6208	0.6135	0.6065	3.4×10^{-11}
Water temperature	−0.0556	−0.0346	−0.0575	9.9×10^{-11}
pH	−0.0352	0.0742	0.0742	1.6×10^{-11}
Specific conductance	−0.0891	0.0907	0.1062	2.2×10^{-11}
Turbidity	0.0228	−0.0486	−0.0360	4.4×10^{-11}

- **Identical models.** All four variants chose the same (weighting, λ, λ_β) in 100% of outer folds for every target. R^2 differs by at most 10^{-10}.
- **Reproduction.** `kron_uncompressed` reproduces the original notebook's saved results (`full_graph_nested_loo_results.csv` and the `Linear + Laplacian` rows of `all_models_node_loo_results.csv`) to within 1.3×10^{-12}.
- **Cost.** Kron took 0.1 s per outer fold vs 15.2 s for the full 67k Laplacian (26 minutes in total), about 150× faster.

When the graph term helps. With chemistry predictors only, the graph term improves pH (+0.11) and specific conductance (+0.18 to +0.20). Both are plausibly controlled by geology and therefore spatially structured. It is neutral or slightly harmful for DO, water temperature and turbidity. Tuning the MERIT weighting instead of using $1/d$ changes R^2 by at most 0.02. The weighting chosen is also unstable from fold to fold for DO (`width` in 13 of 17 folds) and for water temperature (`area` 15, `slope` 5, `area_width` 4 of 24). For pH, `distance_only` was chosen in all 19 folds.

4 Corrections to the original analysis

Table 3: Problems found in `largest_uncompressed_laplacian.ipynb` and how they were handled.

Problem	Fix
pH-transform section used the *unscaled* Kron Laplacian (λ about 21× weaker)	scaled Laplacian; unscaled run alongside for comparison
Static feature sets and Fourier K chosen from outer-LOO results, then re-scored on the same folds	selection moved inside the inner LOO
"Discharge" edge feature is a linear function of log UPA, so it duplicates `area`; its fit also used held-out stations	discharge removed
HAND is 0 at all 26 WQ nodes (river pixels)	HAND removed from the predictors
Comparison tables used hard-coded R^2 values from earlier runs	all tables computed from the run

4.1 pH transforms with the scaled Laplacian

Predictors are specific conductance plus $K = 1$ Fourier terms.

Table 4: pH nested-LOO R^2. The scaling fix changes graph R^2 by only -0.002, because tuning λ over a wide grid absorbs the constant factor.

Target transform	Linear	Graph, scaled (fixed)	Graph, unscaled (original)	Change
raw pH	0.0163	0.1300	0.1315	−0.0016
z-scored pH	0.0163	0.1300	0.1315	−0.0016
$H^+ = 10^{7-\mathrm{pH}}$	−0.0662	0.0930	0.0954	−0.0024

Raw and z-scored pH give identical results, as expected for an affine transform. Fitting H^+ is worse, and no H^+ predictions had to be clipped. Raw pH should be kept.

4.2 Feature selection inside the nested LOO

In each outer fold, the feature set was chosen by inner-LOO MSE alone, separately for the linear and graph models (graph: Kron Laplacians, with weighting, λ and λ_β tuned). The 30 candidates are chemistry plus Fourier $K \in \{1, \ldots, 5\}$, combined with one of six static options: none, one of elevation, log UPA, log width or local slope, or all four. The same run gives each candidate's own nested-LOO R^2. Taking the best of those reproduces what the original analysis effectively reported, so the difference measures how much it was overstated.

Table 5: Outer-LOO R^2 with seasonal and MERIT predictors. "Biased" is the best candidate chosen after seeing outer results. "Fair" is the feature set chosen inside the inner loop, and is the number to report. "Baseline" is $K = 1$ with no static features.

Target	Model	Baseline	Biased	**Fair**	Overstated
Water temperature	linear	0.6781	0.8599	0.8573	0.0026
	graph	0.7394	0.8591	0.8568	0.0024
DO concentration	linear	0.6908	0.7260	0.7109	0.0151
	graph	0.6831	0.7220	0.7009	0.0211
pH	linear	0.0163	0.0268	−0.0553	0.0821
	graph	0.1300	0.1439	0.0977	0.0461
Specific conductance	linear	−0.0793	−0.0564	−0.1690	0.1126
	graph	0.1462	0.2674	0.1403	0.1271
Turbidity	linear	0.0554	0.1954	0.1443	0.0511
	graph	−0.0084	0.1064	0.0740	0.0324

- **Water temperature.** A large and genuine gain, from 0.68 to 0.86, with almost no overstatement. All four MERIT features were chosen in every fold. With them included, the graph term adds nothing, because the node-level features already explain the offsets the Laplacian was capturing.
- **DO.** A modest genuine gain, to 0.70–0.71, driven by $K = 2$–4 seasonal terms with log UPA (graph, 13 of 17 folds) or elevation (linear, 14 of 17 folds). The graph is slightly worse than linear.

- **Specific conductance.** The largest overstatement. Choosing local slope after seeing outer results inflated graph R^2 from a fair 0.14 to 0.27, and the fair value is no better than the plain $K = 1$ baseline (0.146). The graph term is still the only model with positive R^2. Local slope was chosen in 22 of 24 folds, so it may be real, but it does not improve held-out accuracy.
- **pH.** The graph term helps (0.10 vs -0.06 for linear), but R^2 stays low and the chosen features vary from fold to fold.
- **Turbidity.** Linear with $K = 5$ seasonal terms and no static features is best (0.14). The graph term hurts.

5 Reconciliation with the Oct 29 and Nov 5 meeting notes

The model and theory in the meeting notes agree with this report. That includes the objective (1), the $1/n_i$ weighting, the Kron/Schur derivation, preservation of effective resistance, and the compression result. The reported Graph-Laplacian R^2 values differ. Every one of them was copied from an output cell of `largest_uncompressed_laplacian.ipynb`. Table 6 traces each one and gives two corrected values:

- **Benchmark refit:** the graph model fitted inside the updated `water_quality_model_benchmark.py`, on exactly the same rows, predictors and folds as the conventional models.
- **Fair:** the value from Table 5, with features selected inside the inner loop.

Table 6: Graph-Laplacian R^2 as reported in the notes, where each value came from, and the corrected values.

Target	Notes	Source in original notebook	Benchmark refit	Fair
Water temperature	0.8628	cell 118: reduced static set	0.8628	0.8568
DO concentration	0.7261	cell 131: DO base set, K tuned	0.7306	0.7009
Specific conductance	0.2525	cell 143: SC base set, K_{year} tuned	0.2569	0.1403
pH	0.1316	cell 67: unscaled Laplacian	0.1300	0.0977
Turbidity	0.0145	cell 129: other features/rows	0.0264	0.0740

Why the numbers differ.

1. **Post-hoc feature selection.** The Oct 29 notes state that "the outer held-out node is never used for tuning" (§9.1). That holds for λ, λ_β and K, but not for the static feature sets (`REDUCED_STATIC`, `DO_BASE_FEATURES`, `COND_BASE_FEATURES`). These were chosen by comparing outer-LOO R^2 across candidate sets. This accounts for most of the gap between the notes and the fair values, above all for specific conductance ($0.25 \to 0.14$).
2. **The Nov 5 graph rows were not produced by the benchmark.** The original `water_quality_model_benchmark.py` never fitted the graph model. It read its rows from a `graph_laplacian_metrics.csv`, which does not exist, so the values in the notes were entered by hand.
3. **Turbidity was compared on different data.** The graph row was labelled 90,843 rows, but its value (0.0145, RMSE 561.81) comes from a model using specific conductance + pH + K=1 + slope on 88,095 rows. The conventional turbidity models use only slope + K=1 (`TURB_FEATURES`). On those rows and predictors, the graph model scores 0.026.

4. **Fixed vs tuned K.** For DO and specific conductance, the notes' graph values tuned K inside the loop, while the conventional models used fixed K (K=2; K_{day}=1, K_{year}=4). Refitting the graph model with the same fixed K changes R^2 by less than 0.005.

5. **pH scaling.** The notes' 0.1316 used the unscaled Laplacian. The scaled value is 0.1300, a negligible difference.

Updated benchmark. `water_quality_model_benchmark.py` (and the matching notebook) now fits the Graph-Laplacian model inside the same nested LOO as the seven conventional models. It uses the five saved Kron Laplacians, with weighting, λ and λ_β tuned by the same inner-LOO criterion. Fitting the graph model takes about 20 s for all five targets.

Table 7: Nov 5 benchmark: best conventional model vs the Graph-Laplacian model, now fitted on the same rows and predictors. The feature sets are those in `TARGET_CONFIGS`, which were chosen post hoc, so every model's R^2 here is somewhat optimistic.

Target	Best conventional	R^2	Graph-Laplacian	Notes (graph)
Water temperature	Ridge	0.8640	0.8628	0.8628
DO concentration	Gradient Boosting	0.7358	0.7306	0.7261
Specific conductance	Spline GAM	0.0307	0.2569	0.2525
pH	Elastic Net	0.0226	0.1300	0.1316
Turbidity	Gradient Boosting	0.3069	0.0264	0.0145

The rankings in the Nov 5 notes are unchanged. The graph model is best for specific conductance and pH, close to the best for water temperature and DO, and far behind gradient boosting for turbidity. What changes is the size of the advantage once feature selection is done fairly (Table 5). For specific conductance, the graph model's fair R^2 is about 0.14, not 0.25. The conventional models would also need their feature sets selected inside the loop before their numbers can be compared fairly.

6 Conclusions

1. **Kron reduction is exact for this model.** It matches the full 67,401-node Laplacian to rounding error for every weighting, and gives identical hyperparameters and R^2. It is about 150× faster, so the 26×26 Kron matrix should remain the default. Compressing the graph changes nothing either.

2. **The value of the graph term depends on the target.** It clearly helps for pH and, to a lesser extent, specific conductance. For water temperature, DO and turbidity, seasonal and MERIT predictors matter far more than the Laplacian.

3. **Report the fair numbers.** The fair R^2 values in Table 5 should replace earlier figures. In particular, specific conductance should be reported at about 0.14, not 0.22–0.27.

4. **Use the updated benchmark for model comparisons.** The graph model is now fitted inside `water_quality_model_benchmark.py` on the same data as the other models. The hand-entered graph rows in the Nov 5 table should be replaced, in particular turbidity ($0.0145 \rightarrow 0.026$). The next step is to choose the benchmark feature sets inside the inner loop, for all models.

Caveats. With only 17–24 stations, differences in R^2 below roughly 0.03–0.05 are within noise, and no standard errors were computed. For pH and specific conductance the chosen features vary between folds, another sign that those gains are fragile. The daily×annual interaction terms and separate daily and annual K from the original water-temperature section were not among the candidates. In the original they changed predictions by at most 0.009 °C.

Reproducibility. All results come from `largest_uncompressed_laplacian_fixed.ipynb` and are saved in `results/laplacian_fixed/`. The standalone check script is `compare_kron_vs_full_laplacian.py`. The benchmark comparison in Table 7 comes from `water_quality_model_benchmark.py`. The graph rows can be rerun on their own with `BENCHMARK_MODELS=Graph-Laplacian`.